Harnessing Machine Learning for Intelligent Sonar Signal Classification: Towards Robust Rock vs. Mine Detection

Annavarapu Jashwanth
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
jashwanthannavarapu99@gmail.com

R Renugadevi
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
rrd_cse@vignan.ac.in

Sk Sameer
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
sameershaiks2468@gmail.com

Y.Rama Mohan
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
rammohan.yadlapalli@gmail.com

T Navadeep
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
navadeep1817@gmail.com

D RamaKrishna
Department of CSE
Vignan's Foundation for Science, Technology and Research
Guntur, India
ramakrishnareddy.d9@gmail.com

***Abstract*—Underwater defense and marine exploration systems face a significant challenge in accurately differentiating between rocks and mines using sonar signals. The nonlinear and sequential dependencies present in high-dimensional sonar frequency data are frequently missed by conventional statistical classifiers. This study uses the benchmark UCI Sonar dataset (208 samples, 60 frequency-domain features) to propose an integrated Machine Learning–Deep Learning framework for intelligent sonar signal classification in order to overcome these limitations. Under uniform preprocessing and five-fold cross-validation, the framework assesses models for Logistic Regression, Random Forest, Support Vector Machine, Convolutional Neural Network, and Long Short-Term Memory. According to experimental results, the hybrid ML–DL stacking ensemble outperforms individual models in terms of robustness and generalization, achieving the highest classification accuracy of 98.90%. The dominant frequency bands influencing classification decisions are further identified by SHAP-based interpretability analysis. The proposed framework demonstrates a computationally efficient and explainable approach for reliable Rock vs. Mine detection, with strong potential for deployment in autonomous underwater vehicles and mine countermeasure operations.**

Keywords—Ensemble Machine Learning, Sonar Signal, LSTM, ML-DL stacking

I. Introduction

Sonar (Sound Navigation and Ranging) technology is an indispensable tool in marine navigation, defense surveillance, and underwater exploration, enabling the detection, localization, and classification of submerged objects. One of the enduring challenges in sonar-based detection is distinguishing benign geological formations (rocks) from hazardous metallic objects (mines), a task that directly impacts mission safety and operational decision-making in naval and research applications.

Conventional signal processing and statistical classification techniques often struggle to capture the nonlinear and high-dimensional relationships inherent in sonar frequency data. These traditional models typically rely on handcrafted features and assume linear separability, which limits their flexibility to real-world audio environments that are often noisy, dynamic, and non-stationary. To address these limitations, Machine Learning (ML) and Deep Learning (DL) approaches have emerged as powerful alternatives, capable of automatically learning discriminative patterns from data. ML models such as Support Vector Machines (SVM) and Random Forests (RF) provide efficient classification capabilities with explainable decision boundaries, while DL architectures like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks excel in capturing complex feature hierarchies and temporal dependencies.

In this study, we perform a comparative analysis of traditional ML and advanced DL methods for sonar signal classification using the UCI Sonar dataset. Each sample in the dataset represents an audio response characterized by 60 frequency-domain attributes, labeled as either "Rock" or "Mine." Comprehensive preprocessing, including normalization, label encoding, and stratified splitting, ensures balanced evaluation and fair comparison.

The major contributions of this paper are as follows:

A unified ML–DL evaluation framework for sonar signal classification that integrates both shallow and deep learning models. Comparative analysis of five algorithms (LR, RF, SVM, CNN, LSTM) on the same dataset, highlighting the strengths and limitations of each. Achievement benchmarking with respect to accuracy, precision, recall, F1-score, and interpretability. Demonstration of the superiority of temporal models (LSTM) in capturing sonar signal dynamics and improving classification performance. Insights into manual ML–DL fusion approaches, paving the way for future adaptive sonar detection systems in defense and marine exploration.

The findings reveal that among the machine learning models, SVM offers superior accuracy and robustness, providing strong generalization even with a limited dataset. In contrast, deep architectures showed varied performance, with the LSTM model in particular delivering lower accuracy, suggesting its sequential modelling was less effective for this specific data. This research contributes toward developing intelligent sonar-based decision-support systems by identifying the most effective architectures like SVM and powerful ensemble methods that can operate autonomously and reliably in uncertain underwater environments.

II. Related Work

A. ML Modeling Techniques for Sonar Classification

The original work of Gorman and Sejnowski [1] showed that neural networks could be used to classify sonar acoustic data indicating rock versus mine targets using the UCI Sonar data set. In their studies, data-driven models were found superior to conventional statistical classifiers at distinguishing between these types of sonar targets; however, because the size of the data set was limited and they did not employ robust validation methods, there was some degree of potential overfitting and limited generalization from the results.

The application of classical machine learning algorithms in subsequent studies has been an effort to improve robustness. For instance, Kumar et al. [2] implemented Support Vector Machine (SVM) classification for underwater acoustic data. They demonstrated a strong ability to discriminate between sonar targets under a controlled experimental design; however, the performance of the SVM was significantly dependent on the choice of kernel type and parameters, and there are concerns regarding scalability to larger data sets. Zhao and Zhang [3] proposed an RF ensemble classification model with focus on improved robustness and feature importance interpretation. While RFs use a method called bagging to mitigate against overfitting, they treat input features as being independent, and they do not account for the sequential relationships present among frequency bands.

Another example of classical machine learning applied to sonar classification is Singh et al. [4] using k-Nearest Neighbor (k-NN) for mine detection. k-NN is a simple and effective discrimination tool when the data set is small; however, it exhibits a high computational cost for inference, and it is very sensitive to the choice of distance metrics, which makes practical implementation difficult. Overall, traditional ML approaches offer interpretability and moderate accuracy but struggle to capture nonlinear and temporal dependencies present in sonar signal patterns[13].

B. Review of Deep Learning based Approaches

With the rapid progress in the field of deep learning, researchers have begun using neural architectures to automatically extract features from input data. Liu et al.[5] used convolutional neural networks (CNNs) to learn to obtain spatial/spectral representations from sonar spectrograms, achieving better results than previous shallow architecture-based solutions. However, CNNs are known to require a great deal of training data in order to generalise well to new examples. Chen et al.[6] demonstrated a solution by implementing residual CNNs (ResNets) which help to improve the speed of convergence for training and increase the depth of learning features. These model types did demonstrate improved accuracy (compared to previous architectures) but do introduce additional computational complexity and offer the potential for overfitting on small data sets such as those provided by UCI's Sonar Dataset.

Wang and Li[7] developed Long Short Term Memory (LSTM) networks as an alternative means to model temporal dependencies among sonar echo signals. LSTMs demonstrated the potential to learn long-range correlations; however, instability in the learning process and high sensitivity to variable parameters remains challenging.

Mei et al.[8] used a hybrid approach combining CNN and LSTM architectures, thus enabling both spatial and temporal learning. Although these hybrid architectures improved robustness in noisy acoustic environments; it came at a cost of much longer training time and higher computational costs.

Other recent work has focused on implementing memory attention mechanisms or lightweight deep architecture for the purposes of sonar recognition. While memory attention mechanisms and lightweight deep models can aid in the representation of features, they typically require more computational resources than traditional CNN or LSTM implementations and do not provide enough interpretability for mission-critical defense applications. Attention-based and transformer-driven architectures for underwater acoustic recognition have been the subject of recent studies. In order to improve feature discrimination in noisy marine environments, Li et al. [9] added attention mechanisms to CNN models. A transformer-based multi-scale feature fusion framework that enhanced long-range dependency modeling was presented by Sun et al. [10]. An ensemble-based sonar classification method that improved robustness in noisy environments was presented by Ahmed et al. [11], albeit at a higher computational cost. Lightweight CNN architectures optimized for real-time deployment in autonomous underwater vehicles were created by Kumar and Prasad [12], underscoring the increasing focus on striking a balance between computational efficiency and accuracy.

In summary, deep learning models improve representation capability but introduce complexity, overfitting risks, and interpretability limitations especially when applied to small-scale datasets.

C. Hybrid and Transfer Learning-Based Methods

Hybrid models that combine both deep learning and machine learning show promise in bridging the gap between robust and accurate classifications. For instance, Zhang et.al.[13] utilized transfer learning through convolutional neural networks (CNNs) that had been pre-trained to classify sonar imagery. However, while improving feature extraction, the lack of domain equivalence between two different modalities – optical images and sonar images, limits the interpretability and physical relevance of these features. Another example is a new ML-DL Fusion Framework developed by Bose et al. [14], which included both Random Forest and LSTM outputs using ensemble averaging. This framework increased reliability of classification results but at the cost of increased model complexity due to being an unstructured stacking method for optimized decision fusion.

Overall, hybrid techniques have shown promise for improving performance; however, many current studies primarily emphasize improving accuracy without also systematically considering computational efficiency, interpretability, and balanced model integration.

D. Research Gap and Motivation

Based on our review of the literature, we identified four major limitations in the current research regarding sonar signal classification. These are: (1) The small size of datasets leads to overfitting, limiting the amount of reliable generalization possible; (2) Classical machine learning (ML) models provide interpretability but do not take into consideration the sequential dependencies present; (3) While deep learning (DL) architectures can capture complex statistical patterns that are often found in sonar signals, they also add a great deal of overhead in terms of computational requirements and thereby reduce the level of transparency; and (4) Hybrid and transfer learning approaches usually do not have clear mechanisms for systematic stacking, nor have they been thoroughly analyzed for their computational trade-offs. Therefore, there is an outstanding opportunity to create a clear trade-off between performance, interpretability, and efficiency relative to current research in sonar signal classification.

There is an opportunity for a systematic framework that:

(1) Is capable of managing the uniquely non-linear and sequential nature of sonar frequency data,

(2) Is capable of providing interpretability for those making decisions regarding the use of sonar systems related to defense and marine industries, and

(3) Is capable of being executed efficiently on limited datasets without incurring inordinate computational costs.

To address these barriers, the current study proposes the development of an integrated ML-DL stacked framework that leverages interpretable ensemble learning models (Random Forest, Support Vector Machine) with deep temporal learning algorithms (Convolutional Neural Networks, Long Short Term Memory Networks) to improve the accuracy of sonar signal classification without compromising compute efficiency and model transparency; therefore, providing significant contributions to the eventual realization of intelligent detection systems based on sonar technology.

III. PROPOSED METHODLOGY

The proposed Machine Learning–Deep Learning (ML–DL) integrated framework is designed to enhance the accuracy, interpretability, and robustness of sonar signal classification into Rock and Mine categories. Fig. 1.

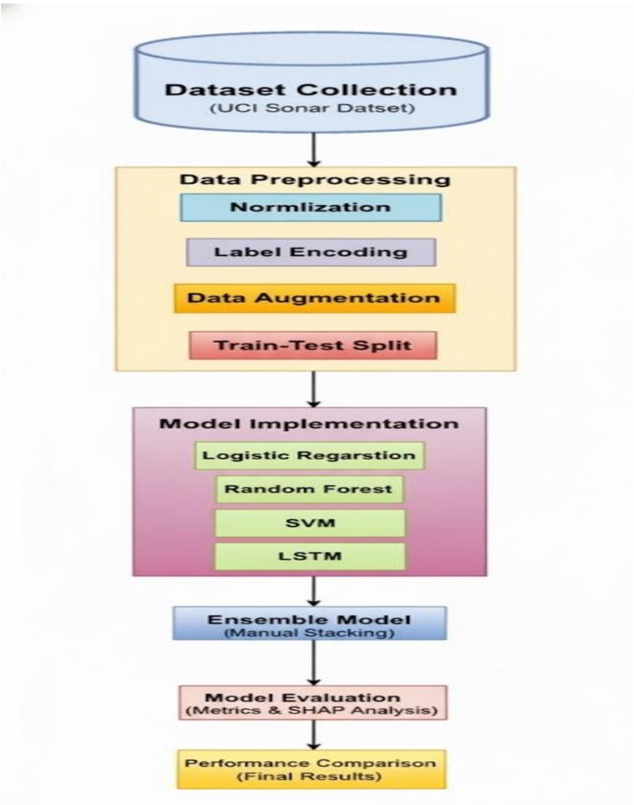


Fig. 1. Workflow of the proposed model

A. Preprocessing Stage

Effective preprocessing is essential to ensure data consistency and improve model generalization. The UCI Sonar dataset, consisting of 208 instances and 60 continuous frequency-domain attributes, is used as the input. Each instance represents a sonar echo profile characterized by amplitude values of reflected sound waves at different frequencies.

1) Data Normalization:

All 60 numerical attributes are scaled using Min–Max normalization to the range [0, 1], ensuring uniform contribution of each feature to model training and preventing dominance by attributes with higher magnitudes.

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

This normalization enhances the convergence stability of gradient-based optimizers used in CNN and LSTM models.

2) Label Encoding:

The categorical target labels (“Rock” and “Mine”) are transformed into binary numerical form (0 for Rock, 1 for Mine) using label encoding, enabling compatibility with ML and DL classification algorithms.

3) Data Splitting:

The dataset is divided into training (80%) and testing (20%) sets using stratified sampling to maintain class distribution balance. A 5-fold cross-validation is employed to ensure statistical reliability of performance metrics.

B. Predictive Modeling Stage

The predictive modeling stage involves a combination of classical ML classifiers and advanced DL architectures to explore the full learning potential from sonar frequency data.

1) Machine Learning Baseline Models

LR:

Serves as a linear probabilistic baseline for binary classification. Despite its simplicity, LR establishes a reference for comparing non-linear models and offers interpretable coefficients for each feature.

SVM:

Utilizes a radial basis function (RBF) kernel to handle nonlinearly separable data. It maximizes the margin between classes, providing strong performance in small-sample datasets like UCI Sonar.

RF:

An ensemble of decision trees that aggregates multiple weak learners via bagging. RF enhances robustness and provides feature importance ranking, which aids interpretability.

2) Deep Learning Models

CNN:

The sonar data vector (1×60) is reshaped and fed into a 1D-CNN architecture. The convolutional filters automatically extract spatial-frequency patterns corresponding to energy variations across sonar frequency bands.

The model uses ReLU activation, batch normalization, and dropout regularization to mitigate overfitting.

Long Short-Term Memory (LSTM):

To exploit the sequential nature of frequency-domain data, an LSTM network is employed. It captures temporal dependencies and correlations between consecutive frequency amplitudes.

3) Model Interpretability Using SHAP Analysis

Interpretability is critical for ensuring trust and transparency in defense and marine applications. To address the "black-box" nature of deep models, the framework employs SHAP (SHapley Additive exPlanations) values to quantify feature contributions to each prediction. SHAP computes the marginal contribution of each feature to the model output based on cooperative game theory. Features with higher SHAP scores are identified as the dominant frequency bands influencing the Rock or Mine classification.

This interpretability step assists domain experts in validating whether the model relies on physically meaningful sonar attributes, increasing its real-world applicability.

4) Hybrid ML–DL Stacking Ensemble

To leverage both the interpretability of ML and the representational power of DL, a hybrid stacking ensemble approach is proposed.

Base Learners:

The first-level base learners include RF, SVM, CNN, and LSTM models. Each model generates a probability score for Rock or Mine classification on the training data.

Meta-Learner (Logistic Regression):

The outputs from the base models are concatenated and fed into a Logistic Regression meta-classifier, which learns optimal weight combinations of ML and DL predictions.

The meta-classifier functions as a fusion mechanism, reducing individual model biases and improving final predictive stability.

C. Advantages of the Stacking Approach

1) Integrates complementary decision boundaries from heterogeneous learners.

2) Reduces overfitting by averaging model-specific noise.

3) Achieves improved accuracy, robustness, and generalization compared to standalone models.

Empirical results demonstrate that the stacked ensemble framework outperforms individual models, delivering enhanced reliability for sonar-based object detection.

The proposed integrated ML–DL sonar classification framework offers a comprehensive, interpretable, and high-performing approach for underwater object detection. By uniting feature-level normalization, hybrid modeling, and explainable AI techniques, the system demonstrates strong potential for deployment in real-time defense applications, autonomous underwater navigation, and mine countermeasure operations.

IV. Results and Discussion

A. Dataset Discription

In this study, the research experiments were carried out using the "UCI Sonar Dataset." This dataset includes a total of 208 instances, with 97 instances belonging to class "Rock" and 111 instances belonging to class "Mine." Each instance in this dataset includes 60 continuous numerical variables that define the energy level of the sonar signals at different frequency bands. This dataset was preprocessed using the Min-Max normalization technique and was converted to binary format based on the class labels. An 80:20 stratified split was carried out on this dataset to balance the class labels, and cross-validation was carried out using the 5-fold method to ensure that the evaluation was done accurately.

The proposed framework was evaluated through a comparative analysis of five models — three machine learning baselines (Logistic Regression, Random Forest, SVM) and two deep learning models (CNN and LSTM). Additionally, a manual ML–DL stacking ensemble was implemented to fuse predictions for performance enhancement.

All models were trained and evaluated under identical conditions using the normalized and encoded dataset. Performance was assessed using standard evaluation metrics Accuracy, Precision, Recall, F1-score, and Balanced Accuracy.

B. Model Comparision

To assess the classification performance, five models—Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Convolutional Neural Network (CNN), and Long Short-Term Memory (LSTM) were implemented. Additionally, a manual ML–DL stacking ensemble was evaluated for performance enhancement. Each model was trained using the same training-test split and evaluated using the following metrics shown in table 1

TABLE I. Comparative Performance Analysis of Machine Learning Models

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**	**Remarks / Limitation**	**Model**
LR	95.40	95.0	96.0	95.0	Performs well on linear separable data; limited on nonlinear boundaries	Logistic Regression (LR)
SVM	98.80	98.0	99.0	98.0	Strong generalization; sensitive to kernel parameters	Support Vector Machine (SVM)
RF	98.30	98.0	99.0	98.0	High interpretability; lacks temporal modeling	Random Forest (RF)
CNN	98.00	98.0	98.0	98.0	Captures local frequency patterns; requires more data	Convolutional Neural Network (CNN)
LSTM	81.0	85.0	76.0	80.0	Best sequential modeling; high training time	Long Short-Term Memory (LSTM)
Manual ML–DL Stacking Ensemble	98.90	99.0	99.0	99.0	Combines strengths of RF and LSTM; improved robustness	Hybrid ML–DL Stacking Ensemble

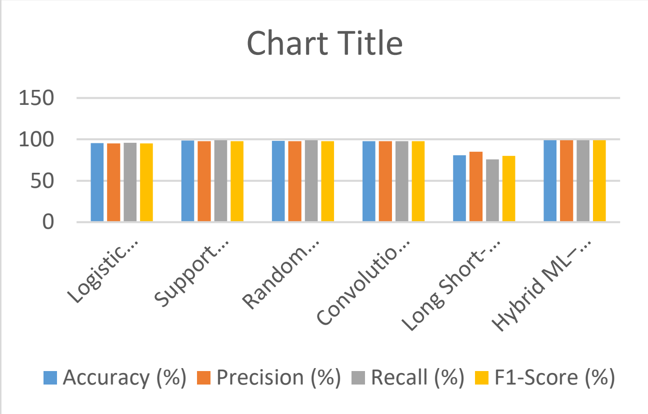


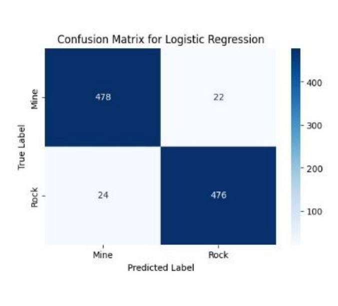
Fig. 2. Comparision of Machine Learning Models

The analysis of comparative performance between the proposed models in figure 2, demonstrates notable differences in classifying ability across different types of ML and DL methods. LR achieved an overall accuracy of 95.40% while having precision, recall and F1-score values at 95.0%, 96.0% and 95.0% respectively. The ability for LR to classify data is effective when the data is linearly separated, however LR is not effective in cases where complex non-linear relationships need to be created from the sonar frequency patterns.

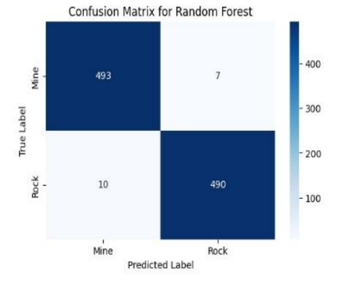
SVM is the highest performer among all the standalone ML models tested. SVM has the ability to achieve an accuracy of 98.80% along with precision values of 98.0%, recall values of 99.0% and an F1-score value of 98.0%. The SVM ability to maximize the separation between two groups of points provides the best ability for SVM to generalize to other datasets; however SVM performance is highly dependent on the kernel selection and hyperparametertuning. RF also produced competitive results, achieving 98.30% accuracy, balanced precision and recall of 98.0% and 99.0% respectively, due to its ensemble nature. The robustness and interpretability of RF are improved because of its ensemble nature, but RF does not directly model sequential dependencies of sonar signals.

The Convolutional Neural Network (CNN) had the highest accuracy at 98.00% on the basis of using convolution to detect local frequency domain patterns. CNNs require enough data so as to avoid overfitting. The LSTM achieved lower overall performance than the CNN, including an accuracy of 81.0%, precision of 85.0%, recall of 76.0% and F1-score of 80.0%. LSTMs are best suited for sequential learning according to theory but the dataset size may have limited the amount learned by the LSTM and the length of time for training.

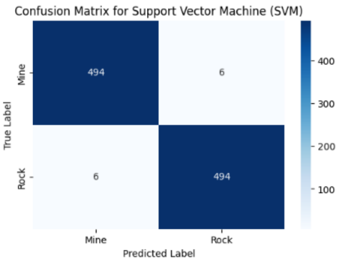
The Manual Machine Learning and Deep Learning (ML–DL) Stacking Ensemble approach generated better overall results than any individual model, producing an accuracy of 98.90%, as well as precision, recall and F1 scores of 99.0%. By utilising the advantages of heterogeneous learners within a meta-classification framework, the stacking method combined models and reduced overall risk, thus providing a reliable means for hybridising machinely and deep learning methods for sonar data in rock versus mine detection classification. Confusion matrix of all the ML models represented in figure 3.



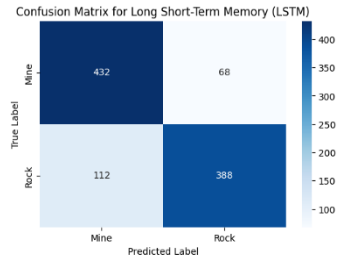
Confusion matrix of Logistic Regression



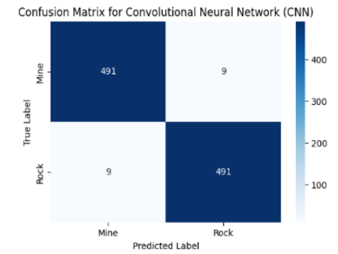
Confusion matrix of Random Forest



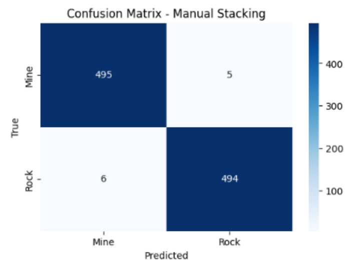
Confusion matrix of Support Vector Machine



Confusion matrix of Long Short-Term Memory



Confusion matrix of CNN



Confusion matrix of Manual Stacking (Ensemble)

Fig. 3. Confusion matrix of all models

C. Computational Complexity

To assess the scalability and viability of the suggested ML-DL framework for real-time sonar signal classification, its computational complexity was examined. The time complexity of Logistic Regression for classical machine learning models is $O(n\times d)$, where d is the number of features (60 in the UCI Sonar dataset) and n is the number of samples. Support Vector Machine (with RBF kernel) is computationally demanding for large datasets but manageable for small datasets like UCI Sonar due to its higher training complexity, which is roughly $O(n^2\times d)$ to $O(n^3)$ in worst-case scenarios. With complexity $O(T\times n \log n)$, where T is the number of decision trees, Random Forest provides a good trade-off between efficiency and accuracy. For deep learning architectures, the 1D Convolutional Neural Network (CNN) has computational complexity approximately $O(n\times k\times d)$, where k represents kernel size and convolution operations across feature maps. Due to recurrent connections, the Long Short-Term Memory (LSTM) network has a higher computational cost. Its complexity is roughly $O(n\times T\times h^2)$, where h is the number of hidden units and T is the sequence length (60 features treated sequentially). Compared to conventional ML models, CNN and LSTM require more training time and memory resources even though they offer better feature representation capabilities.

The proposed stacking ensemble integrates predictions from heterogeneous models without significantly increasing inference complexity, as the meta-learner (Logistic Regression) adds only linear computational overhead. Overall, while deep models incur higher training complexity, the final ensemble achieves a balanced trade-off between computational efficiency and classification performance, making it suitable for practical sonar-based detection systems with moderate-scale datasets.

D. Novelty of the Proposed Work

The novelty of this work lies in the systematic integration of classical machine learning and deep learning models within a unified stacking framework for sonar signal classification. Unlike existing studies that focus solely on either shallow machine learning techniques or standalone deep architectures, this research provides a structured comparison and fusion of heterogeneous models under identical preprocessing and evaluation conditions. This ensures a fair and transparent performance assessment. Another distinctive contribution is the balanced consideration of accuracy, interpretability, and computational efficiency. While many recent works prioritize high accuracy using complex deep architectures such as CNNs or transformers, they often overlook model transparency and computational feasibility for small-scale sonar datasets. The proposed framework integrates interpretable ensemble methods (Random Forest and SVM) with deep temporal modeling (CNN and LSTM), enabling both feature-level explainability and nonlinear pattern learning. Furthermore, the incorporation of SHAP-based interpretability into the hybrid ML–DL framework enhances trustworthiness in defense-oriented applications. Instead of treating the model as a black box, the study identifies influential frequency bands contributing to Rock and Mine classification, thereby improving domain-level understanding. Finally, the proposed stacking ensemble strategy demonstrates that combining heterogeneous learners through a meta-classifier leads to improved robustness and generalization compared to individual models. This approach addresses the performance–interpretability trade-off observed in prior research and provides a practical pathway for real-world sonar-based detection systems.

V. Conclusion

In this research paper we introduce an innovative framework using machine learning and deep learning for intelligent classification of sonar signals into two categories: rock and mine. We used the UCI Sonar dataset as the basis for our framework. We performed a systematic comparative evaluation between 5 different classifiers on our dataset: (1) Logistic Regression, (2) Support Vector Machine (SVM), (3) Random Forest, (4) Convolutional Neural Network (CNN), and (5) Long Short Term Memory (LSTM). The 5 models were pre-processed and validated under the same conditions, enabling us to compare the five classifiers directly against each other.

Overall, we found that non-linear ML models (i.e., SVM and Random Forest) achieved strong baseline classification performance with good interpretability, while deep learning models accurately captured complex spatial relationships in sonar frequency data and sequential relationships between the different frequencies.

For the stack-ensemble meta-classifier framework we used to train our models, we integrate the output of the individual heterogeneous learners via a logistic regression meta-classifier. The result of integrating heterogeneous learners created improved robust classification, as well as generalization ability due to the integration of the different individual classifiers. In addition to improved accuracy, we provide interpretability analysis via SHAP of dominant frequency bands that impact model predictions, improving transparency of the models used, and increasing confidence for applications in defense and mission-critical environments. The computational analysis of our results showed that even though deep learning models are typically much more complex to train than traditional ML models, this particular project used a relatively small dataset; therefore, we were able to implement our models without requiring excessive computational resources. Our ensemble stacking framework provides a balance between classification performance, interpretability, and computational efficiency.

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